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Python Cheat Sheet

Quick-reference syntax for everyday Python development.

DATA TYPES

```
42  # int        Integer
3.14  # float    Float
"hello"  # str   String
True/False       Boolean
[1,2,3]  # list  List (mutable
(1,2)  # tuple   Tuple (immut)
{"a":1}  # dict  Dictionary
{1,2,3}  # set   Set
None  # null     Null value
```

CONTROL FLOW

```
if x > 0:        Condition che
elif x == 0: ..  Else-if branc
else: ...        Fallback bran
for i in range(  For loop
while x > 0: x-  While loop
y="y" if x else  Ternary expr
break / continu  Loop control
pass             No-op placeho
```

FUNCTIONS

```
def greet(n="wo  Default arg
  return f"Hi {  f-string retu
def fn(*args, *  Variadic args
sq = lambda x:   Lambda func
def add(a:int)-  Type hints
@decorator       Apply decorat
yield value      Generator fun
return a, b      Multiple retu
```

LIST TRICKS

```
n[1:3]           Slice [2,3]
n[::-1]          Reverse list
n[-1]            Last element
[x for x in n i  List comprehe
[x**2 for x in   Transform lis
sorted(n, rever  Sort descendi
zip([1,2],['a',  Zip iterables
list(map(fn, ls  Map function
```

CLASSES & OOP

```
class Animal:    Define class
  def __init__(  Constructor
  self.name = n  Instance attr
class Dog(Anima  Inheritance
  def speak(sel  Override meth
    return "Woo  Method body
d = Dog('Rex')   Instantiate
isinstance(d, A  Type check
```

ERROR HANDLING

```
try:             Try block
  res = 10 / x   Risky op
except ZeroDivi  Catch specifi
except Exceptio  Catch any
  print(e)       Log error
finally: cleanu  Always runs
raise MyErr('oo  Raise excepti
with ctx as f:   Context manag
```

FILE & I/O

```
with open("f.tx  Open for read
  text = f.read  Read all text
  lines = f.rea  Read as lines
open("f.txt","w  Open for writ
  f.write("hell  Write string
import json      JSON module
json.dumps(obj)  Object to str
json.loads(s)    Str to object
```

DECORATORS & ASYNC

```
def log(fn):     Define decora
  def wrap(*a,   Wrapper func
    return fn(*  Call original
  return wrap    Return wrappe
@log             Apply decorat
async def fetch  Async functio
  await asyncio  Await corouti
asyncio.run(fet  Run async mai
```

BUILT-INS

```
len(x)  type(x)  Length / type
isinstance(x, i  Type check
range(0, 10, 2)  Range 0,2..8
enumerate(lst)   Index + value
map(fn, lst)     Transform ite
filter(fn, lst)  Filter iter
any([F,T])  all  Any / all tru
max  min  sum    Aggregations
```

STRING METHODS

```
s = " Hello Wor  Define string
s.strip()        Remove whites
s.lower()  s.up  Case convert
s.replace("o","  Find & replac
s.split(" ")     Split to list
",".join(["a","  Join list
s.startswith("H  Prefix check
f'x={x:.2f}'     Format float
```

DICT & SET OPS

```
d = {"a":1, "b"  Create dict
d.get("c", 0)    Safe get
d.keys()  d.val  Keys / values
d.items()        Key-value pai
{"c":3, **d}     Merge dicts
s={1,2}; t={2,3  Create sets
s & t  # inters  Intersection
s | t  s - t     Union / diff
```

GENERATORS

```
def counter(n):  Generator fun
  for i in rang  Iterate range
    yield i      Yield value
g=(x**2 for x i  Gen expressio
next(g)          Get next valu
list(g)          Materialise g
with contextlib  Context manag
  ...            Body block
```